
Machine Learning Based Analog Circuit Fault Diagnosis
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The Biggest Engineering Innovation | Dashboard Implementation Details

PART 1 — The Biggest Engineering Innovation

Chapter 29 — The Topology Rule Part 1 — The Biggest Engineering Innovation (Based on your final dashboard implementation and project evolution)

Before Starting

If someone asks me what makes your project different from 1000 other Machine Learning projects, my answer would not be SVM. It would not be GridSearchCV. It would not be DC Level. It would be this.

The Topology Rule

Because this is where your project stops being "just another ML project" and becomes an Engineering AI System. What Is the Topology Rule? Professor asks What is the Topology Rule?

Perfect Answer

The Topology Rule is a domain-knowledge correction step applied after machine learning prediction. Instead of relying only on the SVM output, the system also considers the physical behavior of the circuit and its topology to resolve specific misclassifications, especially those identified during confusion matrix analysis. Why Was Another Improvement Needed? Question. Didn't DC Level already improve the classifier? Yes. It improved it. But

Machine Learning

still works using statistics. Question. Does

Machine Learning

understand electronics? No. It only understands numbers. The Limitation of Machine Learning Suppose two feature vectors look very similar. The SVM may still predict the wrong class. Question. Can an experienced electronics engineer sometimes recognize the correct fault using physical knowledge? Yes. That is exactly what the Topology Rule tries to capture.

Machine Learning Alone

Pipeline **Waveform** — Features. **SVM** — Prediction. Hybrid System Your project became **Waveform** — Features. SVM

Topology Rule

Final Prediction

Notice something. The SVM is still there. We did not replace it. We improved its decision. Why Is This Powerful? Imagine a doctor. First, the doctor looks at the blood test. Then, the doctor uses experience. Would you prefer a doctor who only reads the blood report? Or one who also uses medical knowledge? Obviously the second. Your project works the same way. AI + Engineering This chapter introduces a very important concept.

Machine Learning

+

Domain Knowledge

=

Hybrid Intelligence

This approach is becoming very common in industry because it often outperforms pure Machine Learning. Why Not Trust the SVM Completely? Question. Is the SVM wrong? No. It is doing its best using the information available. But it has one limitation. It does not know how RC, RLC, or Sallen-Key circuits actually work. You do.

The Topology Rule

adds that missing engineering knowledge.

Real-Life Analogy

Imagine GPS navigation. GPS says

Go Straight

But you see a road is closed. Question. Should you follow the GPS? No. You use human knowledge to correct the route.

The Topology Rule

does exactly that. Where Is It Applied? One important point. The rule is applied after prediction. Not before. Pipeline

Feature Extraction

SVM Prediction

Check Rule

Final Output

Question. Why? Because the SVM should always make the first decision. The rule only intervenes when additional engineering knowledge can improve that decision.

PROFESSOR ASKS

Why is the Topology Rule applied after prediction instead of before training?

EXCELLENT ANSWER

The SVM first performs a data-driven prediction using the extracted features. The Topology Rule is then applied as a post-processing step to incorporate circuit-domain knowledge without changing the trained machine learning model. How Was the Rule Designed? Notice your workflow.

Confusion Matrix

Repeated Errors

Circuit Analysis

Engineering Observation

Topology Rule

The rule did not come from guesswork. It came from evidence. Exactly like DC Level. Why Is This Better Than Retraining? Suppose one specific misclassification keeps occurring. Question. Should we retrain the entire model? Maybe. But if we understand the physical reason, a targeted engineering rule can solve the problem more directly. This makes the system more explainable. Explainability One of the biggest advantages of the Topology Rule is explainability.

Machine Learning

might say **Prediction** — RC_Open. But the rule can explain why the prediction was adjusted. This is important for industrial systems.

Why Companies Like This

Companies do not always want a "black box." They prefer systems whose decisions can be explained. Your project moves in that direction. Instead of blindly trusting the model, it combines data-driven learning with engineering reasoning. This Is Called. In AI, this approach is often called Knowledge-Augmented AI or Hybrid AI You may not have used those exact words

while building

the project, but your design follows the same philosophy.

COMMON MISTAKES

MISTAKE: [X] Thinking the Topology Rule replaces Machine Learning. Wrong. It complements it.

MISTAKE: [X] Thinking rules make Machine Learning unnecessary. Wrong. Rules solve specific cases. Machine Learning handles general classification.

MISTAKE: [X] Adding rules without evidence. Wrong. Rules should be derived from engineering analysis.

REVISION NOTES

Remember the evolution. **Baseline SVM** — GridSearchCV. **Confusion Matrix** — DC Level. **Topology Rule** — Final System. Each step builds on the previous one.

MEMORY TRICK

Remember this sentence. "Machine Learning made the prediction. Engineering made the final decision." That sentence captures the philosophy behind your final system.

KNOWLEDGE CHECK

Level 1

- What is the Topology Rule?
- Why was it introduced?
- Is it applied before or after SVM prediction?
- Does it replace the classifier?
- What is Hybrid AI?

Level 2

- Why doesn't the SVM understand circuit topology?
- How does the Topology Rule improve explainability?
- Why was the rule created from confusion matrix analysis?
- Compare a pure ML system with your hybrid system.
- Why are domain-specific rules useful in engineering applications?

Professor Level

- Explain the complete motivation behind the Topology Rule.
- Why is combining domain knowledge with machine learning often superior to using either alone?
- How does the Topology Rule improve trustworthiness in engineering systems?
- Why is explainability important in industrial AI?
- Explain how your final project evolved from a statistical classifier into a hybrid intelligent system.
- End of Chapter 29 (Part 1)
- Congratulations.
- You now understand the philosophy behind the Topology Rule.
- This is arguably the most mature engineering concept in your project because it demonstrates that you did not rely solely on machine learning—you combined it with domain expertise to build a more reliable and explainable system.

Author's Note The next continuation will cover Chapter 29 (Part 2): The actual implementation of the Topology Rule in your dashboard. Every condition used in the rule. Why those conditions were selected. How the prediction is modified. Real examples showing when the rule activates and when it does not. Professor-level questions specifically about the implementation and its limitations. This chapter will complete the explanation of the final decision-making logic in your project.

PART 2 — Complete Implementation in the Dashboard

Chapter 29 — The Topology Rule Part 2 — Complete Implementation of the Topology Rule (Based on your final dashboard implementation)

Before Starting

Last chapter we answered one question. Why was the Topology Rule created? Today, we answer the next question. How does it actually work? This chapter explains the complete implementation. Where Does the Rule Run? Your final pipeline became Waveform

Feature Extraction

Scaling — SVM Prediction.

Prediction Probability

Topology Rule

Final Prediction

Dashboard Notice something important. The rule does not change the SVM. It only checks its output. Why After Prediction? Question. Why not apply the rule before Machine Learning? Because the SVM should always make the first decision. The rule is a verification layer. Think of it as a second opinion.

Real Life Analogy

Imagine a doctor.

Step 1

Blood Test

Step 2

Doctor Reviews Report

Step 3

Final Diagnosis

Would you remove the blood test? No. Would you remove the doctor? Also no. You need both. Exactly the same idea.

The Dashboard Receives

After prediction, your dashboard has several pieces of information. Conceptually

Predicted Class

Prediction Confidence

Extracted Features

Circuit Type

These are the inputs to the rule.

Step 1 — Check the Circuit

Question. Why? Because different circuits behave differently. Example **RC** — Charging Curve. RLC

Oscillatory Response

One rule cannot fit every topology. That is why the circuit type is checked first.

PROFESSOR ASKS

Why is circuit topology considered?

EXCELLENT ANSWER

Different circuit topologies exhibit different transient characteristics. Applying the same correction rule to every circuit could introduce errors, so the topology is considered before modifying the prediction.

Step 2 — Check the Predicted Class

Suppose the SVM predicts RC_Open Question. Should the rule run for every prediction? No. Rules should only activate for known problematic situations. Otherwise, they may reduce accuracy.

Engineering Principle

A rule should be specific, not universal.

Step 3 — Check Important Features

The dashboard already has the extracted features. Examples DC Level

Rise Time

Peak Energy Question. Why check these? Because they contain physical information about the circuit.

Machine Learning

uses them statistically. The rule uses them physically. Example Suppose SVM predicts RC_Normal But the DC Level clearly matches the expected RC_Open behavior. Question. Should we blindly trust the SVM? No. The rule detects this inconsistency.

Step 4 — Apply the Rule

Now the dashboard asks a simple question. Does the physical behavior agree with the predicted fault? If **Yes** — Keep. the prediction. If **No** — Modify. the prediction. Simple. Decision Flow SVM Prediction

Check Circuit

Check Feature Conditions

Condition Satisfied? Yes

Correct Prediction

No

Keep Original Prediction

Notice something. The rule is conditional. It does not always change the result.

PROFESSOR ASKS
Does the Topology Rule always modify the prediction?
EXCELLENT ANSWER

No. The rule activates only when specific engineering conditions are satisfied. Otherwise, the original machine learning prediction is retained. Why Is This Better? Suppose the SVM is correct 95% of the time. Question. Should the rule change every prediction? Absolutely not. It should only intervene when strong engineering evidence exists. That is exactly how your implementation works. Explainability Suppose someone asks "Why did the final prediction change?" You can answer because the rule detected that the steady-state behavior was inconsistent with the original prediction. Notice how the decision can be explained. That is a huge advantage.

Example Scenario

Imagine these values. **Circuit** — RC. **Prediction** — Normal. DC Level

Matches Open Circuit

Question. What should happen? The rule recognizes the mismatch between the predicted class and the observed steady-state behavior. It then updates the prediction to the physically consistent class.

Why This Is Not Cheating

Some students might think "Isn't this cheating?" No. Because the rule is based on engineering knowledge, not the correct answer. The dashboard never sees the true label. It only sees the measured features and the predicted class. Exactly like a real diagnostic system.

PROFESSOR ASKS

Why is the Topology Rule not considered data leakage?

EXCELLENT ANSWER

The rule operates only on information available during inference, such as the extracted features and predicted class. It never uses the true fault label, so it does not introduce data leakage. Benefits of This Design Your final system has three advantages. 1

Better Accuracy

Specific systematic errors can be corrected. 2

Better Explainability

Every correction has a physical reason. 3

Better Reliability

The system is no longer dependent only on statistical learning. Limitations A good engineer also knows the limitations.

The Topology Rule

is not perfect. It depends on the engineering knowledge used to design it. If new circuit topologies are introduced, the rule may need to be updated. That is completely normal.

PROFESSOR ASKS

What is one limitation of the Topology Rule?

EXCELLENT ANSWER

The rule is based on known circuit behavior. If the system is extended to new circuit topologies or fault mechanisms, the rule may need to be redesigned or expanded.

Why Interviewers Love This

Suppose two students both built an SVM classifier. Student A says "I trained an SVM." Student B says "After identifying systematic misclassifications, I combined machine learning with circuit-domain knowledge through a post-processing Topology Rule that improves explainability and reliability." Who sounds more like an engineer? Obviously Student B. That is the strength of your project. Complete Final Pipeline LTspice Simulation

Dataset Generation

Noise Generation

Feature Extraction (17 Features)

StandardScaler — Optimized SVM.

Confusion Matrix Analysis

DC Level Feature

Topology Rule

Final Dashboard

Fault Diagnosis

This is your complete project.

COMMON MISTAKES

MISTAKE: [X] Applying rules to every prediction. Wrong.

MISTAKE: [X] Designing rules without evidence. Wrong.

MISTAKE: [X] Ignoring physical circuit behavior. Wrong.

MISTAKE: [X] Assuming Machine Learning must always have the final word. Wrong. Hybrid systems often perform better.

REVISION NOTES

Remember the Topology Rule pipeline. **SVM Prediction** — Engineering Check. Condition Satisfied? **Yes** — Correct Prediction. **No** — Keep Prediction. Memory Trick Remember this sentence. "Statistics make the first decision. Physics verifies the final decision." That sentence summarizes the entire Topology Rule.

KNOWLEDGE CHECK

Level 1

- When is the Topology Rule applied?
- Does it replace the SVM?
- Why is the circuit type checked?
- Does the rule modify every prediction?
- Why is the rule explainable?

Level 2

- Explain the complete execution flow of the Topology Rule.
- Why is a hybrid AI system more reliable?
- Why is the rule not considered data leakage?
- What information does the rule use?
- What are the limitations of this approach?

Professor Level

- Explain the implementation of the Topology Rule in your dashboard.
- Why is post-processing preferable to modifying the trained model in this case?
- How does the Topology Rule improve explainability?
- Compare a pure SVM system with your hybrid AI system.
- If you extended this project to amplifier or filter fault diagnosis, how would you redesign the Topology Rule?
- End of Chapter 29
- Congratulations.
- You now fully understand the Topology Rule—both its motivation and its implementation.
- This is one of the strongest contributions in your project because it demonstrates that you didn't stop at achieving good machine learning accuracy. You built a hybrid diagnostic system that combines:
 - Signal Processing
 - Machine Learning
 - Circuit Theory
 - Engineering Reasoning
- That combination is what makes your project stand out during interviews and technical discussions.

Author's Note The next continuation begins Chapter 30 — The Final Dashboard (day17_dashboard.py). We'll explain: The complete GUI architecture. Every button and widget. File upload flow. Model loading (model.pkl, scaler.pkl, label_encoder.pkl). Prediction pipeline inside the dashboard. How results are displayed. Error handling. Why deployment is just as important as model training. This chapter will complete the transformation of your work from a machine learning experiment into a usable software application.